

Multiple Choice (10 marks)

1. Statement 1 | If aH is an element of a factor group, then $|aH|$ divides $|a|$. Statement 2 | If H and K are subgroups of G then HK is a subgroup of G .
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. The set of all real numbers under the usual multiplication operation is not a group since
 - (a) multiplication is not a binary operation
 - (b) multiplication is not associative
 - (c) identity element does not exist
 - (d) zero has no inverse
3. Statement 1 | If H is a subgroup of G and a belongs to G then $|aH| = |Ha|$. Statement 2 | If H is a subgroup of G and a and b belong to G , then aH and Hb are identical or disjoint.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Find all c in $\mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/(x^2 + c)$ is a field.
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
5. Statement 1 | If R is a ring and $f(x)$ and $g(x)$ are in $R[x]$, then $\deg(f(x)+g(x)) = \max(\deg f(x), \deg g(x))$. Statement 2 | If F is a field then every ideal of $F[x]$ is principal.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The characteristic of the ring $2\mathbb{Z}$ is 0, indicating that it has no finite order elements.
7. True or False: A complete graph with 10 vertices has 20 edges, connecting only some of the vertices.
8. True or False: If H and K are subgroups of a group G , then $|HK| = |H||K|/|H \cap K|$ is a true statement.
9. True or False: In the finite field $\mathbb{Z}_2$, the polynomial $x^2 + 1$ has zeros at both $x = 0$ and $x = 1$.
10. True or False: Every group is isomorphic to a subgroup of some group of permutations.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the function $f(x, y) = x^2 - 2xy + y^3$ and discuss the conditions under which its relative extrema occur, particularly focusing on the line $x = y$.
12. Analyze the polynomial function $f(x) = 2x^5 + 8x - 7$ and determine the number of real roots it has. Discuss your reasoning and the methods used in your analysis.

End of Paper